

Algebra II with Trigonometry

Chapter 5.2

Study Guide (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:
[4401268_alg-2trig-h-chp-5-2-note-docx.pdf](#)

Here is a comprehensive study resource based on the provided document.

1. Summary of the Document

This document is a guided lesson on right triangle trigonometry (Chapter 5.2). It introduces the foundational concepts of trigonometry by defining the six trigonometric ratios in relation to the sides of a right triangle. It also covers "special" right triangles (30° - 60° - 90° and 45° - 45° - 90°) to evaluate exact trigonometric values without a calculator. Furthermore, it outlines reciprocal trigonometric relationships and applies these concepts to real-world word problems using angles of elevation and depression.

2. Core Definitions, Theorems, and Formulas

Core Definitions:

- **Solving a Triangle:** The process of determining all of a triangle's unknown parts (the lengths of all three sides and the measures of all three angles) from known information.
- **Line of Sight:** The imaginary line extending from the eye of an observer to the object being viewed.
- **Angle of Elevation:** The angle formed between the horizontal and the line of sight when looking *up* at an object.
- **Angle of Depression:** The angle formed between the horizontal and the line of sight when looking *down* at an object.
- **Angle of Inclination:** The angle of the line of sight when it follows a physical object, like a hillside or inclined plane.

Formulas (Trigonometric Ratios): For an acute angle θ in a right triangle:

- $\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$
- $\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$
- $\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$

Reciprocal Relationships:

- $\csc t = \frac{1}{\sin t}$ (or $\frac{\text{hypotenuse}}{\text{opposite}}$)
- $\sec t = \frac{1}{\cos t}$ (or $\frac{\text{hypotenuse}}{\text{adjacent}}$)
- $\cot t = \frac{1}{\tan t}$ (or $\frac{\text{adjacent}}{\text{opposite}}$)

Special Right Triangles (Derived from the diagrams):

- **45°-45°-90° Triangle:** The ratio of the sides is $1 : 1 : \sqrt{2}$.
- **30°-60°-90° Triangle:** The ratio of the sides is $1 : \sqrt{3} : 2$.

3. Worked Study Guide: Step-by-Step Concepts

Step 1: Finding Missing Ratios in a Right Triangle To find all six trigonometric ratios (like in Examples 1, 2, and 5), you need all three sides of the triangle.

- *Concept:* Use the Pythagorean theorem ($a^2 + b^2 = c^2$, where c is the hypotenuse) to find the missing side.
- *Application:* If you are given $\cos \theta = \frac{12}{13}$ (Example 5a), draw a right triangle. Label the side adjacent to θ as 12, and the hypotenuse as 13. Find the opposite side:
 $12^2 + b^2 = 13^2 \rightarrow 144 + b^2 = 169 \rightarrow b^2 = 25 \rightarrow b = 5$. Now you can find any ratio, e.g., $\sin \theta = \frac{5}{13}$.

Step 2: Using Special Triangles The document requires you to evaluate trig functions for 30°, 45°, and 60° without a calculator.

- *Concept:* Memorize the side ratios of the special triangles shown on Page 1.

- *Application:* To find $\sin(30^\circ)$, look at the 30° - 60° - 90° triangle. The side opposite the 30° angle is 1, and the hypotenuse is 2. Therefore, $\sin(30^\circ) = \frac{1}{2}$. To find $\tan(45^\circ)$, look at the 45° - 45° - 90° triangle. Opposite is 1, adjacent is 1. Therefore, $\tan(45^\circ) = \frac{1}{1} = 1$.

Step 3: Calculating Reciprocals If a calculator only gives you sine, cosine, and tangent, you must use reciprocals for the others.

- *Concept:* Flip the fraction of the base function.
- *Application:* If you need $\csc(48^\circ)$ (Example 3b), you will calculate $\frac{1}{\sin(48^\circ)}$.

Step 4: Setting up Word Problems (Elevation & Depression)

- *Concept:* Always draw a right triangle. The horizontal line is always one of the legs. The "line of sight" is always the hypotenuse.
- *Application:* For an angle of depression (Example 9), draw a horizontal line straight out from the airplane. The angle of depression is between that horizontal line and the downward line of sight. *Note:* Because horizontal lines are parallel, the angle of depression looking down is mathematically equal to the angle of elevation looking up from the ground. Use SOH CAH TOA to write an equation and solve for the missing side.

4. Practice Problems with Answers

These problems are directly adapted from the unworked examples provided in the document.

Problem 1 (Based on Example 1a): A right triangle has a hypotenuse of 41. The side adjacent to angle θ is 40. Find the exact value of $\sin \theta$. >

Answer: First, find the missing opposite side using the Pythagorean theorem: $a^2 + 40^2 = 41^2 \rightarrow a^2 + 1600 = 1681 \rightarrow a^2 = 81 \rightarrow a = 9$. >
Since $\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$, $\sin \theta = \frac{9}{41}$.

Problem 2 (Based on Example 4a): A right triangle has an acute angle of 45° and a hypotenuse of 12. Find the exact length of the adjacent side, labeled x . > **Answer:** Use the cosine ratio:

$\cos(45^\circ) = \frac{\text{adjacent}}{\text{hypotenuse}} \rightarrow \cos(45^\circ) = \frac{x}{12}$. > From the special triangle chart, $\cos(45^\circ) = \frac{1}{\sqrt{2}}$ (or $\frac{\sqrt{2}}{2}$). > $12 \cdot (\frac{\sqrt{2}}{2}) = x$. > $x = 6\sqrt{2}$.

Problem 3 (Based on Example 5c): Sketch a right triangle that has an acute angle θ where $\cot \theta = \frac{5}{3}$. Find the exact value of $\tan \theta$ and $\sin \theta$. >

Answer: Since $\cot \theta = \frac{\text{adjacent}}{\text{opposite}}$, the adjacent side is 5 and the opposite side is 3. > The tangent is the reciprocal of cotangent, so $\tan \theta = \frac{3}{5}$. > To find sine, find the hypotenuse: $3^2 + 5^2 = c^2 \rightarrow 9 + 25 = 34 \rightarrow c = \sqrt{34}$. > $\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{3}{\sqrt{34}}$. (Rationalized: $\frac{3\sqrt{34}}{34}$).

Problem 4 (Based on Example 6a): Evaluate the following expression without using a calculator: $\sin(30^\circ) \cos(60^\circ) + \sin(60^\circ) \cos(30^\circ)$. >

Answer: Using the special triangles provided in the text: > $\sin(30^\circ) = \frac{1}{2}$ > $\cos(60^\circ) = \frac{1}{2}$ > $\sin(60^\circ) = \frac{\sqrt{3}}{2}$ > $\cos(30^\circ) = \frac{\sqrt{3}}{2}$ > Substitute these values in: $(\frac{1}{2})(\frac{1}{2}) + (\frac{\sqrt{3}}{2})(\frac{\sqrt{3}}{2}) = \frac{1}{4} + \frac{3}{4} = \frac{4}{4} = 1$.

Problem 5 (Based on Example 8): The angle of elevation to the top of the Empire State Building is found to be 11° from the ground at a distance of 1 mile from the base of the building. Write a trigonometric equation to find the height (h) of the building in miles. > **Answer:** The distance along the ground is the adjacent side (1 mile), and the height of the building is the opposite side (h). The correct ratio connecting opposite and adjacent is tangent. > $\tan(11^\circ) = \frac{h}{1}$ > $h = \tan(11^\circ)$ **miles**.

5. Assumptions and Ambiguities Resolved

- **Blank Worksheets as Source Material:** The provided document is largely a blank worksheet (notes meant to be filled out during a lecture). Because the prompt requires me to base the output *only* on the document, I assumed the intended goal of the study guide and practice problems was to successfully complete the blank examples (Examples 1-11) using the concepts introduced on pages 1-3.
- **The Pythagorean Theorem:** The document never explicitly prints the formula $a^2 + b^2 = c^2$. However, solving Examples 1, 2, and 5 is impossible

without it. I assumed foundational knowledge of the Pythagorean theorem is expected to find missing sides of right triangles before applying the trig ratios.

- **Consecutive Mileposts:** In Example 10, the text mentions viewing "two consecutive mileposts." I assumed the mathematical interpretation of this phrasing means the distance between the two points is exactly 1 mile.
- **Units of Measurement:** In Example 8, the distance is given in miles, but building heights are traditionally measured in feet. Because no conversion factor ($1 \text{ mile} = 5280 \text{ feet}$) is provided in the text, I assumed calculating the answer in miles (as done in Practice Problem 5) is perfectly acceptable based strictly on the provided material.